Introduction to Rings Exercises

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July 2025

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1 Question-57

1.1 Problem Statement

Suppose that R is a ring with no zero-divisors and that R contains a nonzero element b such that $b^2 = b$. Show that b is the unity for R.

1.2 Solution

$b^2 = b \Rightarrow b(b-1) = 0$ But as $b \neq 0$ hence $b - 1 = 0$ as R has no zero divisors, thus $b = 1$.

But this solution is wrong as R has a unity is not given. We have to prove also that. So, correct and direct approach that $\Rightarrow$

Let $r \in R$ be any element in R. Then we have to show $rb = r = br, \forall r \in R$.

Now, $rb = rb^2$ as $b = b^2$, so we have:

$$rb = rb^2$$

$$\Rightarrow rb - rb^2 = 0$$

$$\Rightarrow (r - rb)b = 0$$

Since R has no zero-divisors, either $r - rb = 0$ or $b = 0$. But $b \neq 0$ as given, hence $r - rb = 0$ or $rb = r$. Thus, $rb = r$ for all $r \in R$. Hence b is the unity for R.

2 Question-58

2.1 Problem Statement

Find an example of a commutative ring R with unity suhc that $a, b \in R, a \neq b, a^n = b^n$, and $a^m = b^m$, where m and n are psitive integers that ajre relajtively prime.

2.2 Solution

Take $\mathbb{Z}_4$. This is a commujtative ring with unity. Now, let $a = 2$ and $b = 0$. Then we have:

$$a^2 = 2^2 = 0 \mod 4$$

$$b^2 = 0^2 = 0 \mod 4$$

Thus, $a^2 = b^2$ and

$$a^3 = 2^3 = 0 \mod 4$$

$$b^3 = 0^3 = 0 \mod 4$$

Thus, $a^3 = b^3$. Here, $m = 2$ and $n = 3$ are relatively prime.

3 Question-55

3.1 Problem Statement

Let R be a commutative ring with more than one element. Prove that if for every nonzero element a of R we have $aR = R$, then R has a unity and every nonzero element has an inverse.

3.2 Solution

As $aR = R \forall a \neq 0, a \in R$ then there is some element $i_a \in R, i_a \neq 0$ such that $ai_a = a = i_a a$.Now, our goal is to show that i_a doesnt depend on the choice of a. Let $b \in R, b \neq 0$. Now, we have:

$$ab = ai_a b$$

$$\Rightarrow a(b - i_a b) = 0$$

Now, given $a \neq 0$. if $b - i_a b \neq 0$ also, then a has a zero devisor. But if so, then consider $(b - i_a b) = x$ then by the given condition $xR = R$. So, $ar = axr' = 0 \forall r \in R$ as $\forall r \in R \exists r'$ such that $xr' = r$. Contradicting the fact that $aR = R$. Hence $bi_a = b \forall b \in R$. Hence, i_a is the unity of R.
Now, we have to show that every nonzero element has an inverse. That is now same kind of argument as above.
As $aR = R$ meaning $\exists a' \in R$ such that $aa' = 1 = a'a$.

4 Question-54

4.1 Problem Statement

Show that $4x^2 + 6x + 3$ is a unit in $\mathbb{Z}_8[x]$

4.2 Solution

$(4x^2 + 6x + 3)^2 = 16x^4 + 36x^2 + 9 + 2(24x^3 + 18x + 12x^2) = 4x^2 + 1 \mod 8$
Now, $(4x^2 + 1)^2 = 16x^4 + 8x^2 + 1 = 1 \mod 8$

Thus,$(4x^2 + 6x + 3)^4 = (4x^2 + 6x + 3) \cdot (4x^2 + 6x + 3)^3 = 1$

Thus, $4x^2 + 6x + 3$ is a unit in $\mathbb{Z}_8[x]$.

5 Question-53

5.1 Problem Statement

Let R and S be commutative rings.Prove that (a, b) is a zero-divisor in $R \oplus S$ iff a or b is a zero-divisor or exactly one of them is zero.

5.2 Solution

Let (a, b) is a zero divisor. Then $(a, b) \neq 0$ and there is $(c, d) \in R \oplus S$ such that $(c, d) \neq 0$ and $(a, b)(c, d) = (ac, bd) = 0 \Rightarrow ac = 0, bd = 0 \Rightarrow$ either $a = 0$ or $c = 0$ and $b = 0$ or $d = 0$. But if $a = 0, b \neq 0 \Rightarrow d = 0 \Rightarrow c \neq 0$ and vice-versa.
So either $a = 0$ or $b = 0$

On the other hand, if $a \neq 0$ and $c \neq 0$ then $ac = 0 \Rightarrow a$ is zero divisor. and $b \neq 0$ and $d \neq 0$ then b is zero divisor.

6 Question-52

6.1 Problem Statement

If a, b and c are elements of a ring, does the equation $ax + b = c$ always have a solution? If it does, mujst the solution be unique? Answer the same questions given that a is a unit.

6.2 Solution

No, the equation $ax + b = c$ does not always have a solution. For example, in the ring $\mathbb{Z}_4$, if we take $a = 2$, $b = 0$, and $c = 3$, then there is no x such that $2x + 0 = 3$. If solution exists even then it does not necessarily ujnique. For example, in the same ring thake $a = 2, b = 0, c = 0$ then $2x = 0$ has two solutions $x = 0$ and $x = 2$.
If a is a unit, then the equation $ax + b = c$ always has a unique solution. Since a is a unit, there exists an inverse a^{-1} such that $aa^{-1} = 1$. We can rearrange the equation as follows:

$$ax = c - b$$

Multiplying both sides by a^{-1} gives:

$$x = a^{-1}(c - b)$$

If there is another solution x', then we have:

$$ax' = c - b$$

Subtracting the two equations gives:

$$a(x - x') = 0$$

Since a is a unit, it has no zero divisors, so we can conclude that $x = x'$. Thus, the solution is unique.

7 Question-51

7.1 Problem Statement

Give an example of Boolean ring with four elements.Give an example of an infinite Boolean ring.

7.2 Solution

A Boolean ring is a ring in which every element is idempotent, meaning $x^2 = x$ for all x in the ring. An example of a Boolean ring with four elements is $\mathbb{Z}_2 \times \mathbb{Z}_2$. The elements of this ring are $(0,0)$, $(0,1)$, $(1,0)$, and $(1,1)$. The addition and multiplication operations are defined component-wise. For example:

$$(0,1) + (1,0) = (1,1)$$

$$(0,1) \cdot (1,0) = (0,0)$$

In this ring, every element is idempotent:

$$(0,0)^2 = (0,0)$$

$$(0,1)^2 = (0,1)$$
$$(1,0)^2 = (1,0)$$
$$(1,1)^2 = (1,1)$$

An example of an infinite Boolean ring is the power set of a set, which is the set of all subsets of that set. For example, let S be an infinite set, such as the natural numbers $\mathbb{N}$. The power set $\mathcal{P}(S)$ is an infinite Boolean ring where the elements are subsets of S. The addition operation is symmetric difference, and the multiplication operation is intersection. In this ring, every element (subset) is idempotent because for any subset $A \subseteq S$, we have $A \cap A = A$.

8 Question-50

8.1 Problem Statement

A boolean ring is commutative.

8.2 Solution

A Boolean ring is indeed commutative. In a Boolean ring, every element x satisfies the property $x^2 = x$. This implies that for any two elements a and b in the ring, we have:

$$(a+b)^2 = a^2 + ab + ba + b^2$$

Since $a^2 = a$ and $b^2 = b$, we can rewrite this as:

$$(a+b)^2 = a + ab + ba + b$$

$$a + b + ab + ba = (a+b)^2 = (a+b)$$

$$ab = -ba$$

But $abab = (ab)^2 = ab = (-ba)(-ba) = (ba)^2 = ba$ Thus, $ab = ba$, which shows that multiplication in a Boolean ring is commutative. Therefore, every Boolean ring is commutative.

9 Question-49

9.1 Problem Statement

Let R be a ring. Prove that $a^2 - b^2 = (a+b)(a-b) \forall a, b \in R$ iff R is commutative.

9.2 Solution

$$(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2 \Rightarrow ab = ba$$

Hence, R is commutative.

10 Question-48

10.1 Problem Statement

Determine the smallest subring of $\mathbb{Q}$ that contains $\frac{2}{3}$.

10.2 Solution

The smallest subring is $\mathbb{Z}[\frac{2}{3}] = \{a_0 + a_1\frac{2}{3} + \cdots + a_n\frac{2^n}{3^n} : n \in \mathbb{N} \text{ and } a_i \in \{0,1,2\}\}$.